THE RATTY2TAILS PAPERS

MONETARY HISTORY SERIES



The Birth of ₿itcoin

How a thirty-year hunt for private digital cash ended in a Genesis Block, and what has happened since

EDITED BY RATTY2TAILS · OCTOBER 2026

NOT AN ORIGINAL WORK. This essay is an edited retelling and update of "The Quest For Digital Cash" by Alex Gladstein, published by Bitcoin Magazine on 13 October 2021. The research, reporting and narrative thread belong to the original author. Please read the original.

SOURCE NOTICE

This essay is based on an existing article and is not an original piece of work. The history told in Parts I to V follows the story set out in *"The Quest For Digital Cash"* by **Alex Gladstein**, first published by *Bitcoin Magazine* on **13 October 2021** (page last modified October 2025).[1] Gladstein did the reporting and chose the thread that ties these events together: Adam Back, the cypherpunks, and the long search for digital cash.

What Ratty2Tails has done: condensed and reordered the material, rewritten it in this series' own words, added two diagrams, and written Part VI, which brings the story forward from the article's date to October 2026. **What Ratty2Tails has not done:** carried out new primary research on the pre-2009 history. For that, go to the original article and the primary sources it links.

Dating note: the article carries a 2021 publication date. Its "today" figures (a price above $55,000 and a market value above $1 trillion) describe October 2021, and Part VI compares them with October 2026. This paper is for education and is not financial advice.

TL;DR

- **Bitcoin is the last piece of a long puzzle, not a bolt from the blue.** From the 1970s, a community of cryptographers, the *cypherpunks*, tried to build private digital cash.
- **Two problems kept blocking them.** Every early system needed a central operator who could be shut down, bribed or corrupted, and nobody could make strangers agree on a ledger without a referee.
- **Nakamoto's 2008 design answered both** with a public ledger kept by thousands of volunteers and with proof-of-work borrowed from Adam Back's Hashcash. Back was the first person Nakamoto wrote to.
- **The Genesis Block carries a political message** about bank bailouts. Together with the 21-million cap, it shows the project was about money as much as privacy.
- **Since 2021:** a third and a fourth halving cycle began, spot ETFs arrived, a record high of about $126,000 came in October 2025, and a drawdown of roughly half followed. The US and Europe diverged on state digital money. A quantum-computing debate opened. And in April 2026 a New York Times investigation named Back as its leading Satoshi candidate. He denies it.

INTRODUCTION AND SCOPE

Why this story still matters

Most people meet bitcoin as a price: a number on a screen that is either exciting or alarming. This paper steps back to the moment before the price existed, when bitcoin was a rumour on a mailing list for privacy obsessives. The question is simple. *Why did it take until 2008 to build digital cash, and why did it work when everything before it failed?*

The answer is a relay race. Public-key cryptography handed privacy to ordinary people. David Chaum showed that money could be made private. Adam Back, Wei Dai, Nick Szabo and Hal Finney each supplied a part that Chaum's centralised design lacked. Satoshi Nakamoto, whoever they are, joined the parts together, added a difficulty adjustment and a hard cap, and published it.

Scope. Parts I to V cover the history, roughly 1976 to 2013, drawing on Gladstein's account. Part VI is new and covers October 2021 to October 2026: market data, issuance, institutions, state money, the identity question and the quantum debate. It is not a price forecast. Where a figure comes from a news aggregator rather than a primary source, the reference list says so. Please check a live source before relying on any number.

PART I

The privacy itch

The story starts with a worry rather than a coin. By the early 1990s the internet was spreading, and so was the realisation that every connected life leaves a record that governments and companies can collect, join together and hand over. In 1983 the journalist David Burnham had warned of a "computer state" and urged citizens to demand legal protection. A small group of programmers did not want to wait for legislation. They wanted a technology the state could not switch off.

The technology already existed. In 1975 and 1976, Whitfield Diffie, Martin Hellman and Ralph Merkle worked out how two strangers could exchange private messages without first sharing a secret. Each person publishes a key that locks messages and keeps a second key that unlocks them. Anyone can write to Alice; only Alice can read. A patent followed in 1977, and the NSA's early attempts to discourage publication failed once activists photocopied the paper and passed it around. Cryptography, until then a military preserve, was out.

In 1992 John Gilmore, Eric Hughes and Timothy May began meeting in San Francisco and started a mailing list. This was "The List", the home of the movement and, sixteen years later, the place where the bitcoin white paper would appear. Hughes's 1993 manifesto gave them a slogan: *cypherpunks write code*. Their reasoning was that lobbying changes laws slowly and reversibly, while shipping software changes what is possible at once.

Adam Back, a British computer scientist who had first studied economics as a teenager and later took a doctorate in distributed systems, was among the List's busiest contributors. He got his chance to test the theory in the **Crypto Wars**. Washington treated strong encryption as a munition and tried to restrict its export. Phil Zimmermann's free email-encryption program, PGP, became the test case, with Hal Finney among its early helpers. The government investigated Zimmermann, floated the "Clipper chip" backdoor, and argued that encryption served criminals. The cypherpunks answered by printing the source code in books and shipping them abroad, on the theory that no administration would ban a book. Back sold T-shirts carrying a few lines of code. By 1996 the case against Zimmermann was dropped, and court rulings began treating code as protected speech. PGP became the most widely used email encryption in the world.

> *The authorities had tried to stop the code, and the code had won. Back carried that lesson into 2008.*
> EDITORIAL PARAPHRASE OF GLADSTEIN'S FRAMING

PART II

Cash with a landlord

Having won the argument over private messages, the community turned to private payments. Science fiction fed the appetite: Neal Stephenson's novels imagined worlds of digital cash. A real prototype was already underway. In the early 1980s the Berkeley cryptographer **David Chaum** invented *blind signatures*, a way for a bank to certify a digital token without learning who later spends it. In 1985 he published a paper arguing that this kind of payment could make "Big Brother" obsolete. In 1989 he moved to Amsterdam and founded **DigiCash**, which let people convert ordinary money into electronic tokens that could be sent without leaving a trail.

The cypherpunks loved the privacy and distrusted the architecture. DigiCash's tokens behaved like cash, but the company sat in the middle. It had to run the mint, it could be pressured by regulators, it could go bust, and it could in principle decide how much money existed. It did go bust, later in the 1990s, short of funding. Back drew the lesson that mattered: **digital cash would have to be decentralised, with no single point of failure.**

Back knew the costs of centralisation first-hand. He once ran a "mixmaster" anonymous remailer from a friend's server in Switzerland, paying the rent with cash in the post to avoid leaving a trail. When Swiss police called on his friend, he shut it down the next day. The dream of digital cash outlasted the server.

PART III

The missing pieces

Privacy was only half of what a decentralised system needed. The other half was agreement. If thousands of strangers each keep a copy of a ledger and nobody is in charge, how do they agree which payments are genuine and stop one coin being spent twice? Computer scientists call this

family of problems the **Byzantine Generals Problem**: how do separated commanders coordinate an attack when some messengers may be traitors? Before 2008, every working e-cash system sidestepped the question by appointing an administrator.

The pieces that eventually let Nakamoto skip the administrator arrived one at a time, all through the same mailing list and the same small circle:

- **Hashcash (1997, Adam Back).** An anti-spam stamp: to send a message you first spend a little computing effort that anyone can cheaply verify. This is *proof of work*, and it makes "costly to create, easy to check" a property of digital objects. Nakamoto cited it in the white paper. As Aaron van Wirdum has put it, it let Bitcoin both prevent double-spending without a referee and issue new coins without an issuer.
- **b-money (1998, Wei Dai).** A scheme for pseudonymous participants to pay each other and enforce agreements with no outside help, drawing on Back's proof of work. It was never practical, but its ideas echoed through the List.
- **Bit gold (1998, Nick Szabo).** The big conceptual step: try to make the "provable costliness" of gold digital, so that a token has value because creating it demonstrably took effort. It tied the cypherpunks' privacy dream to the older argument for hard money. It was never built.
- **Reusable proof of work, RPOW (2004, Hal Finney).** A working system that took the bit-gold idea and added servers to verify it. It ran on Finney's own machine, with a plan to decentralise later.

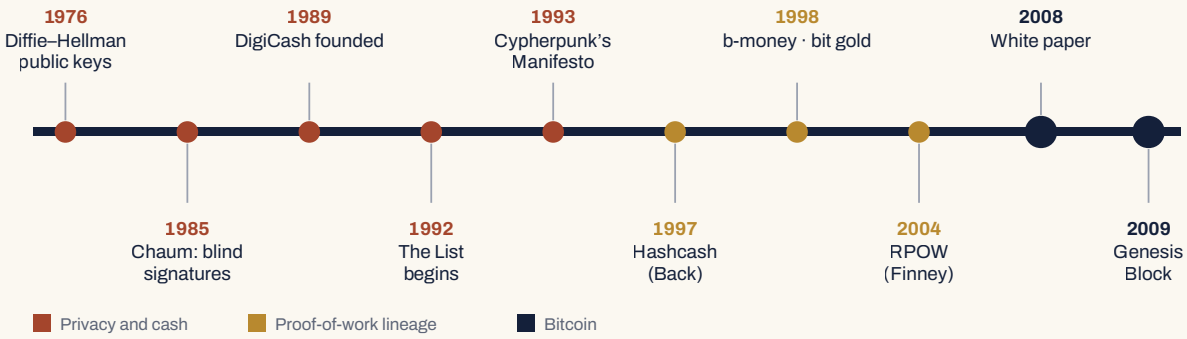


FIG. 1 · The relay race to Bitcoin, 1976–2009. Events are evenly spaced for legibility, not drawn to a time scale. Compiled by Ratty2Tails from events described in the source article.

PART IV

August 2008 to January 2009

In August 2008 Adam Back received an email from a person calling themselves Satoshi Nakamoto, the first time Nakamoto reached out to anyone about the project. Back, intrigued, replied with pointers to related work, including Wei Dai's b-money. On 31 October 2008 Na-

kamoto posted the white paper to the List under the subject line “Bitcoin P2P e-cash paper”, describing an electronic cash system that was fully peer-to-peer and needed no trusted third party. It cited Hashcash and b-money.

The design in one paragraph: instead of one bank’s database, thousands of participants each hold a full, continually updated copy of every transaction ever made (Nakamoto’s own early word for the structure was a *timechain*). If someone tries to spend the same coin twice, everyone else can see it and rejects the second attempt. New coins are earned by the people who add the next page of records, and adding a page requires burning real computing effort, which is Hashcash doing the job Back designed it for. The software went live on **9 January 2009**. The next day Hal Finney, who had been waiting for exactly this, posted two words: “Running bitcoin.”

The Genesis Block

The first record on the chain contains a line of text that Nakamoto chose: *“The Times 03/Jan/2009 Chancellor on brink of second bailout for banks.”* It is a newspaper headline, a timestamp and a statement of purpose. Central banks were creating money to prop up failing institutions. Nakamoto’s later writing was blunt about the root problem: conventional money requires enormous trust, and the history of fiat currency is full of broken trust. Bitcoin was presented as an alternative that replaced discretion with rules.

This is a point Gladstein emphasises. Before the late 1990s, most cypherpunks cared far more about privacy than about monetary policy. The fusion of the two, private transactions on one hand and a supply nobody can inflate on the other, is what made Bitcoin different from DigiCash. Digital rights groups that defend privacy to this day have often failed to see why fixed issuance matters to that goal.

Engineered scarcity and a vanishing founder

Two design features impressed Back most. The first was the **difficulty adjustment**: every 2,016 blocks, about two weeks, the network recalibrates how hard mining is, so that blocks keep arriving about every ten minutes whether miners join in droves or flee. This repaired the weakness he had seen in Hashcash, where faster computers could swamp a currency built on it. The second was the **supply schedule**: a block subsidy that halves roughly every four years and a ceiling of just under 21 million coins.

The third feature was social. In early 2011 Nakamoto stopped posting and has never been conclusively heard from since. For Back this made the project stronger, because a leader who can be arrested or pressured is a point of failure, and a leader who vanishes is not.

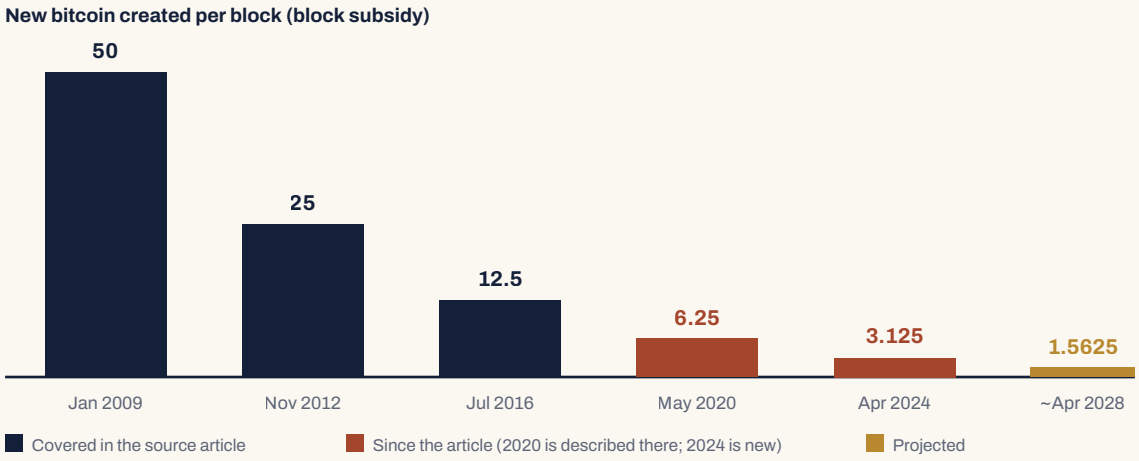


FIG. 2 · The halving staircase. Each step halves issuance. The 2028 date is a projection that moves with block times.[4]

PART V

The privacy gap and Back's quest

Bitcoin, as launched, gave up a lot of privacy to gain decentralisation. A public ledger is what lets any home user audit the system, but it also lets an observer follow money once an address is linked to a person. Back thinks this was the right engineering call. More private designs existed on paper, including zero-knowledge schemes, but they made each transaction far larger and slower, and a system that only data centres can verify is a system that can be captured. Cheap verification keeps ordinary people in charge of the rules, which is the point of the "don't trust, verify" mantra.

By 2013 Back judged that Bitcoin was stable enough to build on, and he threw himself into improving it. One suggestion was to swap the signature scheme from ECDSA to **Schnorr signatures**, which Nakamoto had avoided because of a patent that had since expired. That change arrived with the **Taproot** upgrade, which activated in November 2021, a month after the article was published, as it anticipated.[1] A second idea, **Confidential Transactions**, would hide amounts while still letting everyone check that no coins were created from thin air. Getting it into Bitcoin proved hard, so Back helped found **Blockstream** in 2014 with a seed round of about $21 million, with Greg Maxwell and Austin Hill. The company built a testbed that became the **Liquid** sidechain, where confidential transactions run today, and it invests in the **Lightning Network**, a second layer for cheap, fast payments. The article also records the Blocksize War of 2015 to 2017, in which users resisted pressure to enlarge blocks and so keep node costs low.

The vision, in Gladstein's telling, is a Bitcoin that offers the privacy of paper cash and the savings properties of gold. He adds an urgent reason to care: governments were beginning to experiment with central bank digital currencies, which could be monitored, taxed automatically and even programmed to expire.

PART VI · NEW MATERIAL BY RATTY2TAILS

Since the article: October 2021 to October 2026

Everything in this part is outside the source article and is the editor's own compilation from the sources listed at the end. Treat it with the usual caution applied to fast-moving markets.

A. Then and now

Item	October 2021 (article date)	October 2026
Price of one bitcoin	Above $55,000	About $82,000–$85,000 in early October 2026 (aggregator reports; check a live quote)[5]
Record high	Not stated	About $126,000, reached on 6 October 2025[3]
Cycle low since the record	—	About $58,500 in June 2026, a fall of roughly 52%[3]
Coins mined	Nearly 19 million	20 million passed on 9 March 2026 (block 940,000); under 1 million remain[4]
Block subsidy	6.25 BTC	3.125 BTC since the halving of 20 April 2024; next halving projected around April 2028[4]
Signatures / privacy upgrade	Taproot "next month"	Taproot active since November 2021
US spot bitcoin ETFs	No US spot ETF approved yet	Launched January 2024; about $150bn in assets at the October 2025 peak, about $103bn in late September 2026[3]

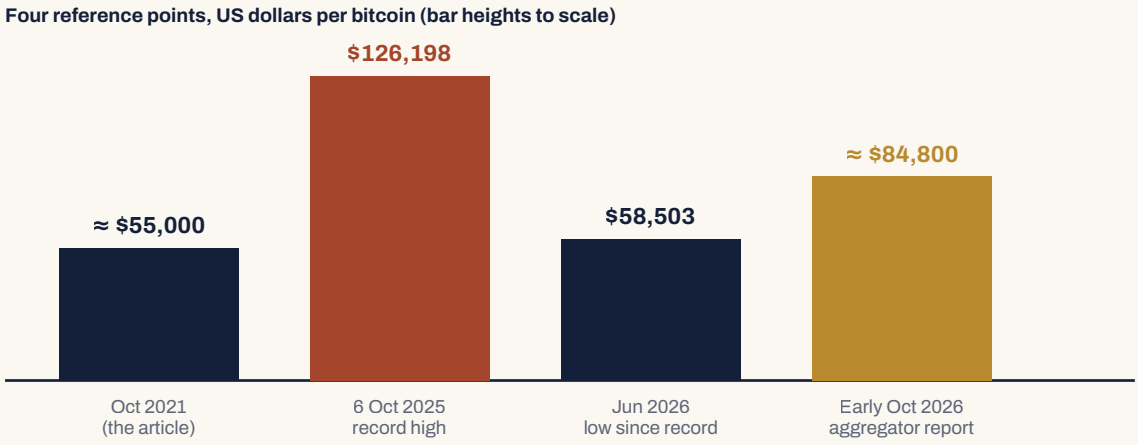


FIG. 3 · Four reference points, not a price series. The 2021 value is "above $55,000" in the article; the 2026 values come from the sources cited.[1,3,5]

B. Wall Street arrives, then the drawdown

The article noted that bitcoin had been adopted by Wall Street and some governments. That process went further than most would have guessed in 2021. US spot bitcoin ETFs launched in January 2024 and drew in money fast: by October 2025 BlackRock's fund was nearing $100 billion in assets much sooner than the largest S&P 500 ETF managed.[6] A US **Strategic Bitcoin Reserve**, set up by executive order in March 2025, gave the 2025 rally a state-backed story. By late 2026 the US government formally recognised a stockpile of about 328,000 BTC, largely seized coins, but commentators note that it has not been buying more.[3,10] The record of about $126,000 came on 6 October 2025. Days later, a leveraged liquidation event wiped out roughly $19 billion of value, and over the next eight months the price roughly halved to a June 2026 low of about $58,500 before recovering into the mid-$80,000s.[3,6] Some analysts argue that the old four-year halving-driven cycle has weakened in favour of demand waves from institutions. That is an opinion, not a fact. Past patterns do not guarantee future ones.

Two observations are worth making in the spirit of the article. First, Back's fixed-supply thesis was tested hard: the asset's dollar price is volatile, but its issuance schedule did not bend. Second, the pre-2021 failure mode the article warns about, central operators, did not vanish; it reappeared in the industry around bitcoin. The collapse of the exchange FTX in November 2022 is the best-known example of customers trusting a middleman with coins they never held in their own keys.

C. The state answers: stablecoins in America, CBDCs in Europe

The article closed with a warning about central bank digital currencies. The picture in 2026 is split. In the United States, an executive order in January 2025 barred federal agencies from developing a CBDC, the House passed an Anti-CBDC bill in July 2025 (219 to 210) that has stalled in the Senate, and the **GENIUS Act**, signed in July 2025, put regulated private stablecoins in the role of America's digital dollar.[8] The Bank of Japan's deputy governor summed up the divergence in May 2026: the US has steered toward stablecoins and Europe toward a CBDC. The ECB plans a pilot in 2027 and issuance no earlier than 2029, with holding limits.[8] China's e-CNY has grown large and, from 1 January 2026, pays interest.[8]

For this essay's theme the lesson is mixed. Cypherpunk worries about surveillance money are live policy debates, not paranoia. But America's answer to surveillance money is a dollar token issued by licensed companies, which is still a trusted third party, and not bitcoin. Bearer digital cash with no issuer remains a narrow, unusual category, and bitcoin is its main member.

D. Who is Satoshi? The spotlight lands on Back

The article treats Nakamoto as a pseudonym whose disappearance strengthened the system. Since then the identity question has returned with unusual force and unusual attention on the article's own protagonist. In October 2024 the HBO documentary *Money Electric* speculated that the developer Peter Todd was Nakamoto, which Todd rejected as "ludicrous", though it also spent much of its time on Back.[7] In April 2026 *The New York Times*, in an investigation by John Carreyrou and Dylan Freedman, named **Adam Back** as its leading candidate, pointing to writing style and early forum activity. Back replied that the report is interpretation, not cryptographic proof, and that it is "definitely not me."[7]

There is no cryptographic proof in the public record that settles the matter: the clean test would be moving coins from the earliest addresses, and that has not happened. It is a good reminder of why this paper's own source is careful to describe Back as the first person Nakamoto wrote to, and nothing more. Whoever Nakamoto is, Bitcoin has now run for over seventeen years without them.

E. A new threat model: quantum computing

The cryptography that Diffie and Hellman unlocked in 1976 and that Bitcoin's signatures depend on is the same cryptography a sufficiently large quantum computer could eventually break. In 2026 the subject moved from theoretical to practical in industry discussion. Proposals such as **BIP-360** (a new output type that reduces how often public keys are exposed) and the more aggressive **BIP-361** (a phased sunset of quantum-vulnerable outputs) are under discussion, and on 13 October 2026 Coinbase urged developers to begin a multi-year transition.[9] No quantum upgrade has been approved or activated. Post-quantum signatures are much larger than today's, which collides with the same "keep it cheap to verify" principle Back defended. Expect this to be the defining technical argument of the coming years.

F. What has not changed

Confidential Transactions, Back's signature idea, are still not part of Bitcoin's base layer to the best of the editor's knowledge; they live on Liquid. The cap of 21 million, the difficulty adjustment and the ten-minute rhythm are untouched. And the founding wager remains what it was in 2008: that code, widely distributed, can do things that law and goodwill cannot.

SUMMARY

What the story teaches

Bitcoin's birth was a relay, not a flash of genius. Public-key cryptography (1976) gave ordinary people privacy; Chaum (1980s) showed money could be private but left it in a company's hands; Back, Dai, Szabo and Finney (1997–2004) built proof-of-work money in pieces; and in 2008–2009 Nakamoto solved agreement without a referee, added a hard cap, and left. The Genesis Block's headline shows the project was always as much about money as about privacy.

Since the source article appeared, the price has made a record and a deep drawdown, the asset has been wrapped in ETFs and recognised in a US reserve, states have chosen different digital money paths, Back himself has been named in the press as a Satoshi suspect, and quantum computing has become a live protocol question. The article's central claim, that decentralised digital cash is hard and worth defending, has not been refuted by any of this. It has simply become a much bigger conversation. — *Ratty2Tails*

Editor's notes on the source

- The article says the subsidy fell to 6.25 BTC in **April 2020**. The halving occurred in **May 2020**; Fig. 2 uses May.
- The Byzantine Generals metaphor is about the Byzantine Empire. The article's mention of the Ottoman Empire in the same sentence blurs two different eras, so this paper omits it.
- Where the article's figures describe October 2021 (price, market cap, number of users), they are used here only as the "then" column in Part VI.
- This paper does not reproduce the article's text. Its quotations from primary documents are limited to the Genesis Block message and a few words from Nakamoto, Finney and Back.

REFERENCES

Sources

1. **Primary source for this essay.** Alex Gladstein, "The Quest For Digital Cash", Bitcoin Magazine, 13 October 2021 (modified 1 October 2025). bitcoinmagazine.com/culture/bitcoin-adam-back-and-digital-cash
2. Satoshi Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System", 31 October 2008. bitcoin.org/bitcoin.pdf (cited in the article)
3. DataWallet, "Bitcoin (BTC) Statistics & Trends for 2026" (price history, ETF assets, US reserve holdings). datawallet.com/crypto/bitcoin-statistics. See also CryptoPotato, "The 2025 Bitcoin Report: Strategic Reserves, Record Highs, and Q4 Crash".
4. Forklog, "Fewer than 1 million bitcoins remain to be mined" (March 2026); Mugglehead, "Bitcoin supply passes 20 million milestone" (halving of 20 April 2024; projected 2028 halving); Bitcoin.com News on the 2028 halving.
5. Pluang / CryptoTicker market note, 4 October 2026 (price near $84,800). Aggregator source; verify against a live quote.
6. CryptoPotato and Coinspeaker coverage of the October 2025 liquidation event and record high.
7. Time, "Has the Mystery of Bitcoin's Creator Been Solved?" (Money Electric, 2024); Bloomberg, "Adam Back Denies He Is Anonymous Bitcoin Founder Satoshi Nakamoto", 9 April 2026; IDN Financials summary of the New York Times investigation, April 2026.
8. Ryozo Himino (Bank of Japan), "Singleness of money and the role of central banks", BIS, 16 May 2026; wiki.private.law, "CBDCs in 2026"; eco.com CBDC update, May 2026 (e-CNY interest from 1 January 2026).
9. Coinbase quantum-security plan, 13 October 2026 (via KuCoin Blog); The Block on BIP-361; BIP-360 coverage (Bitcoin Magazine, Quantum Insider, March 2026).
10. The Motley Fool via Yahoo Finance, 6 March 2026, on the Strategic Bitcoin Reserve not yet buying new coins.

Prepared for education only; not investment advice. Figures from market aggregators are approximate. This edited essay is a derivative commentary on the article cited above and does not replace it.